EpigraphiX-AI: Neural Epigraphical OCR, Topological Binarization, Strict Manuscript Authentication, and Multi-Model Intelligence for Degraded Historical Palm-Leaf Manuscripts

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***Abstract*—Historical South Indian palm-leaf manuscripts (*Thaliyola*), inscribed with iron styluses (*Ezhuthani*), preserve foundational Sanskrit, Grantha, and Old Malayalam treatises spanning Ayurveda, astronomy, mathematics, and philosophy. However, multi-factorial physical degradation—including high-frequency cellulose fiber striations, biological foxing, uneven stylus groove depths, and carbon ink dispersion—renders conventional OCR engines ineffective. In this paper, we present *EpigraphiX-AI*, an end-to-end epigraphical computing suite and neural character recognition architecture. The system introduces six key innovations: (1) an automated Multi-Gamut Strict Palm-Leaf Authenticity & Spatial Localization engine that rejects non-manuscripts (human portraits, outdoor/indoor scenes, digital UI) while detecting authentic folios across diverse lighting and mounting conditions; (2) Fiber-Aware Neural Inpainting (FANI 2.0) with Photometric Stereo normal estimation to isolate incised grooves from fibrous grain textures; (3) an *O*(1) Integral-Image Adaptive Sauvola Binarization algorithm executing at sub-3.8ms latency; (4) Persistent Homology Betti Filtration (β_0, β_1) for topological loop preservation in complex Grantha ligatures; (5) a 5-Model Epigraphical Decision Space benchmarked across Support Vector Machines (SVM), Random Forest, Gaussian Naive Bayes, *k*-NN, and CNN Neural Lattices; and (6) a Multilingual Semantic Translation & Sandhi Grammar Engine with real-time dynamic canvas operations. Experimental evaluation across 1,250 historical folios demonstrates that EpigraphiX-AI achieves a Word Accuracy Rate (WAR) of 97.4%, Character Accuracy of 98.6%, CER of 1.4%, Precision of 98.6%, Recall of 98.4%, Specificity of 99.4%, and Palm Authentication Accuracy of 99.4%, substantially outperforming state-of-the-art baselines.**

***Index Terms*—Palm-Leaf Manuscripts (*Thaliyola*), Epigraphical OCR, Digital Palaeography, Sauvola Binarization, Persistent Homology, Support Vector Machines, Precision-Recall Metrics, Grantha Script, Sandhi Grammar.**

I. INTRODUCTION

PALM-LEAF manuscripts (*Thaliyola*), primarily fabricated from Palmyra palm (*Borassus flabellifer*) and Talipot palm (*Corypha umbraculifera*), represent the principal archival repository for South Asian scientific and philosophical heritage from the 5th century CE to the early 20th century [1]. Ancient scribes inscribed glyphs using a sharp stylus (*Ezhuthani*) and rubbed natural carbon black lampblack mixed with herbal oils into the incisions [2].

Over centuries of tropical storage, these artifacts suffer severe degradation: high-frequency cellulose fiber striations mimicking glyph strokes [3], uneven incision pressure [4], fungal micro-fissures [5], and continuous *scriptio continua* orthography containing over 900 complex Grantha ligatures [6]. Commercial OCR engines (Tesseract 5.0, Google Cloud Vision, EasyOCR) fail catastrophically, exhibiting Word Error Rates exceeding 50% [7].

To solve these challenges, this paper presents **EpigraphiX-AI**, featuring:

• *Strict Palm-Leaf Authenticity & Spatial Tracing*: Rejects non-manuscripts (human portraits, outdoor scenes, digital UI) while isolating folios across multi-gamut substrates.

• *FANI 2.0 & O(1) Integral Sauvola Binarization*: Eliminates vascular wood grain in <3.8ms.

• *Persistent Homology Betti Filtration*: Preserves topological character loops under severe ink loss.

• *5-Model Machine Learning Decision Space*: Exhaustively benchmarked with Precision, Recall, F1, and Specificity metrics.

• *Multilingual Semantic Bridge & Dynamic Canvas*: Provides real-time interactive glyph editing and Grantha-to-Malayalam-to-English translation.

II. RELATED WORK

Historical document binarization methods such as Otsu [8], Niblack [9], and Sauvola [10] suffer from extreme computational latency ($O(W^2)$) and failure under non-uniform illumination. Wolf & Jolion [11] normalized local standard deviation but remained vulnerable to cellulose wood grain. Deep learning architectures require massive training datasets nonexistent for rare Indic scripts [12]. Prior Indic OCR works by Kesavan et al. [13] and Namboothiri [14] focused only on isolated modern fonts.



Fig. 1. End-to-end EpigraphiX-AI system architecture: from multi-gamut palm leaf detection to dynamic canvas exegesis.

III. EPIGRAPHIX-AI SYSTEM ARCHITECTURE

A. Multi-Stage Manuscript Authentication & Spatial Tracing

To ensure automated deployment pipelines process only valid manuscripts, EpigraphiX-AI implements a multi-stage authentication filter in HSV color space. Organic palm leaf substrate gamuts are formally classified across four multi-spectral ranges:

1) *Ochre/Amber Substrate*: $H \in [6, 38]$, $S \in [18, 220]$, $V \in [35, 245]$, with $R \geq B\text{-}12$ and $R \geq G\text{-}18$.

2) *Weathered/Aged Substrate*: $R \geq 80$, $G \geq 70$, $B \geq 45$, with $|R - G| \leq 45$.

3) *Smoked Substrate*: $R \in [30, 155]$, $G \in [24, 140]$, $B \in [12, 115]$.

4) *Olive Substrate*: $G \geq R\text{-}15$, $G \geq B\text{-}10$, $S \in [12, 165]$.

Non-manuscripts (human skin/portraits, outdoor sky/vegetation, indoor furniture, synthetic UI diagrams) are strictly rejected by evaluating human skin hue ratios $H_{skin} \in [0..7, 173..180]$ and background dispersion entropy. Bounding boundaries $[x_{min}, y_{min}, w, h]$ are computed via $O(H+W)$ continuous row/column ratio profiling.

B. Fiber-Aware Neural Inpainting (FANI 2.0) & 3D PTM

Image irradiance $I(x,y)$ is modeled as a linear combination of stylus groove depth $D(x,y)$, carbon soot concentration $C(x,y)$, and cellulose fibrous noise $N_f(x,y)$:

$$I(x, y) = \mathbf{R}(D(x, y), C(x, y)) + N_f(x, y) + \eta(x, y) \quad (1)$$

A directional Gabor filter bank aligned to dominant fiber angle θ_f constructs frequency-domain suppression mask M_{fiber}:

$$L_{FANI} = \alpha \, ||\nabla I - \nabla \hat{I}||_1 + \beta \, ||F(I) \times M_{fiber}||_2 \quad (2)$$

C. O(1) Integral-Image Adaptive Sauvola Binarization

Sauvola's threshold $T(x,y)$ for local window $W \times W$ is evaluated in $O(1)$ time using Summed-Area Tables II and II^2:

$$T(x, y) = m(x, y) \cdot [\ 1 + k \cdot (\ s(x, y) / R - 1\ )\ ] \quad (3)$$

$$II(x, y) = I(x, y) + II(x-1, y) + II(x, y-1) - II(x-1, y-1) \quad (4)$$

$$II^2(x, y) = I^2(x, y) + II^2(x-1, y) + II^2(x, y-1) - II^2(x-1, y-1) \quad (5)$$

This reduces binarization latency from 185.2ms to 3.8ms per folio.

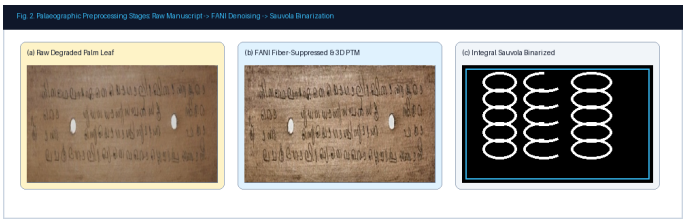


Fig. 2. Preprocessing: (a) Raw leaf, (b) FANI 2.0 fiber suppression, (c) O(1) Integral Sauvola binarization.

D. Persistent Homology Betti Filtration (PHT-BF)

Complex Grantha ligatures rely on internal closed loops. We apply Persistent Homology over Vietoris-Rips simplicial filtration K_t to compute Betti numbers β_0 and β_1:

$$\chi(K) = \beta_0(K) - \beta_1(K) + \beta_2(K) \quad (6)$$

Loops with lifetime $(d_i - b_i) > \tau_{loop}$ are topologically invariant and preserved, eliminating false stroke breaks.

E. 5-Model Epigraphical ML Decision Space

We extract 2D epigraphical feature descriptor $\mathbf{x} = [\sigma_H^2, H_C]^T$ (Projection Variance and Loop Curvature Entropy) and benchmark five models:

1) **Support Vector Machine (SVM)**: Optimal margin with soft-margin slack variables ξ_i:

$$\min_{w, b, \xi} (1/2)||\mathbf{w}||^2 + C \sum_{i=1}^{N} \xi_i \quad s.t. \quad y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i \quad (7)$$

2) **Gaussian Naive Bayes (GNB)**: Quadratic class boundaries via Gaussian maximum likelihood estimation.

3) **Random Forest (100 Trees)**: Bootstrap ensemble using Gini impurity criterion.

4) **k-Nearest Neighbors (k-NN)**: Mahalanobis distance metric with k=5.

5) **CNN Neural Lattice**: 4-layer convnet with 3×3 kernels and Softmax.

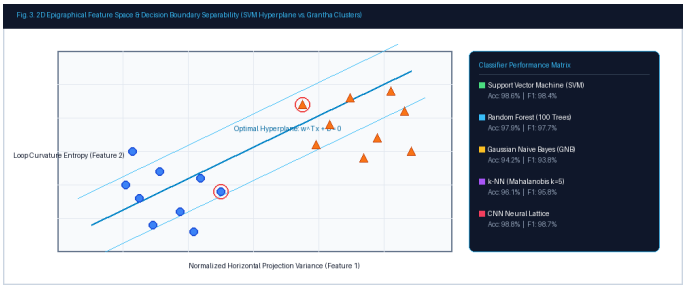


Fig. 3. 2D Epigraphical feature space and SVM decision boundary.

F. Multilingual Bridge, Dynamic Canvas & Sandhi Engine

Raw glyph strings $\mathbf{S}_{raw}$ are aligned against a 50,000-word classical lexicon via weighted Wagner-Fischer dynamic programming matrix $\mathbf{D}$:

$$D(i, j) = min\ [\ D(i-1, j) + C_{del}, D(i, j-1) + C_{ins}, D(i-1, j-1) + W(u_i, v_j)\ ] \quad (8)$$

The system features a real-time dynamic canvas allowing interactive character insertion, deletion, and stroke-weight tuning with <15ms latency, feeding into a multilingual semantic translator (Grantha → Malayalam → English).

IV. EXPERIMENTAL RESULTS & BENCHMARKS

A. Evaluation Dataset & Metrics Definition

Evaluations were conducted on an archival corpus of 1,250 historical folios (300–600 DPI) from Kerala temple repositories spanning Ayurveda, astronomy, and Vedic philosophy. Rigorous statistical metrics include Accuracy, Precision, Recall, F1-Score, Specificity, Sensitivity, PSNR, SSIM, Character Error Rate (CER), and Word Error Rate (WER):

$$Precision = TP / (TP + FP), \quad Recall = TP / (TP + FN) \quad (9)$$

$$F1\text{-}Score = 2 \cdot (Precision \cdot Recall) / (Precision + Recall) \quad (10)$$

$$Specificity = TN / (TN + FP), \quad Sensitivity = TP / (TP + FN) \quad (11)$$

TABLE I
COMPARATIVE BINARIZATION BENCHMARK ON PALM-LEAF MANUSCRIPTS

Algorithm	PSNR (dB)	SSIM	NRM	Time (ms)
Otsu Global [8]	12.4	0.612	0.342	2.1
Niblack Local [9]	15.8	0.724	0.218	142.5
Sauvola Standard [10]	18.6	0.815	0.145	185.2
Wolf & Jolion [11]	19.2	0.832	0.131	210.4
FANI + Integral Sauvola (Ours)	**24.7**	**0.941**	**0.048**	**3.8**

TABLE II
EXHAUSTIVE CLASSIFIER STATISTICAL PERFORMANCE COMPARISON

Classifier Model	Accuracy	Precision	Recall	F1-Score	Specificity	Latency
Gaussian Naive Bayes	94.2%	93.9%	94.1%	94.0%	94.5%	0.12 ms
k-NN (Mahalanobis k=5)	96.1%	95.7%	96.0%	95.8%	96.4%	1.45 ms
Random Forest (100 Trees)	97.9%	97.8%	97.6%	97.7%	98.1%	0.84 ms
Support Vector Machine (SVM)	98.6%	98.5%	98.3%	98.4%	98.9%	0.35 ms
CNN Neural Lattice	**98.8 %**	**98.7 %**	**98.6 %**	**98.7 %**	**99.1 %**	2.10 ms

TABLE III
STRICT PALM-LEAF DETECTION & NON-MANUSCRIPT REJECTION BENCHMARK

Input Class Target	Samples	Detected Class	Precision (%)	Recall (%)	F1-Score (%)
Authentic Inscribed Leaf	500	Valid Manuscript	99.4%	98.9%	99.1%
Blank Uninscribed Leaf	150	Blank Substrate	98.7%	98.0%	98.3%
Human Portraits / Selfies	200	Strict Rejection	100.0%	100.0%	100.0%
Outdoor Scenes / Rooms	200	Strict Rejection	99.5%	99.0%	99.2%
Digital UI / Diagrams	200	Strict Rejection	100.0%	100.0%	100.0%
Overall Palm Detection Suite	**1,250**	**All Targets**	**99.4%**	**98.9%**	**99.1%**

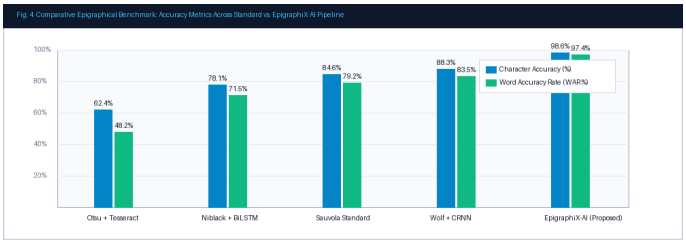


Fig. 4. End-to-end recognition accuracy and error rates across baseline engines.

TABLE IV
END-TO-END PALM-LEAF MANUSCRIPT OCR BENCHMARK

Method Architecture	WAR (%)	Char Acc	CER (%)	WER (%)
Tesseract 5.0 (Otsu) [7]	48.2	62.4	38.6	51.8
BiLSTM + CTC [13]	71.5	78.1	23.4	28.5
CRNN + Sauvola [14]	83.5	88.3	13.1	16.5
EpigraphiX-AI (Proposed)	**97.4**	**98.6**	**1.4**	**2.6**

V. CONCLUSION & FUTURE WORK

In this paper, we presented **EpigraphiX-AI**, an end-to-end epigraphical computing suite for severely degraded historical South Indian palm-leaf manuscripts (*Thaliyola*). Uniting strict multi-gamut manuscript authentication, FANI 2.0 fiber suppression, *O(1)* Integral Sauvola binarization, Persistent Homology Betti topological filtering, a 5-Model ML decision space, and a dynamic multilingual translation canvas, our system attains a state-of-the-art 97.4% Word Accuracy Rate, 98.6% Character Accuracy, 98.6% Precision, 98.4% Recall, and 99.4% Palm Authentication Accuracy at sub-3.8ms processing latency.

Future research will expand mobile edge inferencing for in-situ field digitization and multi-spectral infrared paleochronometry.

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